

March 14, 2025

Subject: Veriten Response to “Request for Information on the Development of an Artificial Intelligence (AI) Action Plan”

Reference: 90 FR 9088, Document Number 2025-02305

Veriten appreciates the opportunity to submit feedback to the Office of Science and Technology Policy (OSTP) in response to its Request for Information on the Development of an Artificial Intelligence (AI) Action Plan.

AI has the potential to create the next industrial revolution by unlocking existing natural resources deemed previously non-economic, enabling rapid prototyping of new industrial processes in virtual labs, strengthening domestic semiconductor manufacturing and supply chains, creating new carbon composite materials at scale to replace heavier metal and cement products, developing new compounds to nurture our crops, and by creating new AI models to enhance and analyze existing data sources. Our recommendations for the Action Plan are designed to be immediately actionable and enable the long-term development of impactful AI applications. Energy is both a source to enable AI, and a key area of policy focus today as noted below, but government policy must also enable the use of AI in the energy sector to ensure U.S. global leadership in AI.

Veriten is an energy research, investing, and advisory firm that draws from its energy-focused network and knowledge platform to bring diverse perspectives to the long-term energy discussion, helping industry partners and investors improve strategic decision-making through the firm’s three lines of business.

1. Through public content, Veriten aims to educate, inform and collaborate with energy producers, energy consumers, academics, politicians, policymakers, and other open-minded individuals and organizations.
2. Veriten serves as a trusted “sparring” partner across confidential and complex situations to 30+ partner companies spanning the entire energy value chain including oil & gas, power, infrastructure, mining & minerals, industrials, financial, policy, and other strategic areas.
3. Veriten applies the analysis from its research and advisory work to its NexTen fund’s investment strategy. The NexTen fund leverages Veriten’s broad network and expertise to invest in reliable, sustainable, scalable, and economically viable energy solutions. Dedicated to understanding the constantly evolving, volatile, and often noisy and confusing energy landscape, the firm supports twelve innovative, private companies poised to capitalize on long-term, global energy trends. Each of these companies are uniquely utilizing AI to enhance their products and services.

In 2025, Veriten launched a series of invitation-only Chatham House rules AI Roundtables. The roundtable, consisting of a group of CEOs, board members, and industry luminaries representing 20+ companies across the energy landscape, is focused on identifying transformational applications for AI within the energy space as well as the unique barriers to effective adoption of existing AI technologies. Our focus includes data quality and availability, effective safety and governance structures, human capital considerations, supportive policies and regulations, strategies for return on investment evaluation, and other key stakeholder considerations and decisions.

Our comments and recommendations reflect insights gained from our research and discussions with a cross section of participants in the roundtable, however, this document represents the sole views of the authors listed in this submission and may not be attributed to any other party.

ENSURING SUFFICIENT ENERGY TO POWER AI AND ACHIEVE U.S. AI & ENERGY DOMINANCE

Energy dominance is a strategic priority of the Trump administration. Energy Secretary Chris Wright and Interior Secretary Doug Burgum recently emphasized that energy is fundamental to the economy and essential for sustaining U.S. manufacturing and AI innovation, explicitly linking energy and AI strategies. Each begets the other but together are of paramount importance to sovereignty and prolonged supremacy for the next century.

The energy sector is keenly aware of its complex supply chain with a unique perspective on local, federal and global pressures. Energy companies live in this physical reality and understand limiting factors of the infrastructure that enables social and economic prosperity. Virtual bits must flow over tangible cables and lines. Data and processing in the cloud resides in data centers that require power and steel in the ground. We believe policymakers shaping AI strategy can gain valuable insights from the energy and industrial sectors, especially amid evolving trade and technological competition with China. Veriten operates within this dynamic landscape, analyzing how shifting priorities could disrupt business models.

To address the unprecedented surge in electricity demand driven by AI and data centers, we advocate for the following key policy actions that will strengthen the U.S. energy framework.

- **Reliable and Affordable Energy** – Uninterrupted, low-cost, baseload energy is vital across the AI supply chain, from domestic chip manufacturing to powering AI model training and inference. The private sector is swiftly mobilizing to expand energy capacity for data centers, with Hyperscalers announcing annual

investments of \$234B to \$249B in 2025 and 2026¹—strong market signals for new generation and transmission infrastructure. However, Gartner predicts that 40% of data centers will face operational constraints due to power shortages.² Some Hyperscalers are even revising their spending plans downward, not from lack of enthusiasm, but from lack of capacity to absorb their demand.³ Meanwhile, NERC and industry groups from both the technology and power sectors emphasize the urgent need for permitting reforms to accelerate new capacity development and meet surging demand.⁴

- **Technoeconomic Forecasting** – The federal government should conduct comprehensive analyses to forecast the grid impacts of various approaches to data center and AI-driven demand growth. Factors such as grid resilience, emissions, and additionality introduce complexity and uncertainty for developers and utilities. Balancing data center flexibility, increased generation capacity, clean energy mandates, and efficiency improvements presents overlapping challenges. A strategic combination of these levers will likely offer the most cost-effective and rapid path forward. Rigorous study of their interactions and limitations can provide critical insights for setting informed national energy targets.
- **Dispatchable Generation Build Out** – Hyperscalers are still defining their long-term energy strategies including the mix of nuclear, renewables, natural gas with carbon capture, geothermal and other power sources. In the near term, leveraging natural gas-fired cogeneration in industrial hubs like Texas⁵ and Louisiana could accelerate capacity expansion while permitting reforms navigate federal, state, and local regulatory hurdles. Since demand for grid-based power exceeds historical buildout speeds by 5x in even the most conservative projections,⁶ utilities, data centers, and major energy companies must collaborate to find or create substantially greater capacity than the grid can provide. This must be accomplished while avoiding at least three major constraints: (i) political fallout from increased retail power costs (ii) reduced energy capacity for America-

¹ [Visual Capital Projection of Hyperscaler Spend in 2025 and 2026](#)
[How AI Energy Requirements are Shaping Data Center Investment](#)

² [Gartner Fall Report 2024 Report on Data Center Growth](#)

³ [Microsoft Revises \\$80 Billion Data Center Spend Down, Jan 2025](#)

⁴ [2024 NERC Long Term Reliability Assessment](#)
[Explosive Demand Growth in Data Center and Reliability Challenges](#)
[NERC Large Loads Task Force \(LLTF\)](#)

⁵ On January 28, 2025, Chevron announced a partnership with Engine 1 and GE Vernova to power U.S. data centers – [Chevron press release January 28, 2025](#)

⁶ Exhibit 17 and 18 in [Goldman Sachs Report on AI, data centers and the coming US power demand surge](#)

First industrialization onshoring / reshoring efforts (iii) managing emissions commitments.

- **Prolonging Existing Incentives** – Tax credits and loan programs have significantly influenced developer strategies, particularly for nuclear energy projects. Re-commissioning retired nuclear plants was once unthinkable, and interest in expanding existing nuclear capacity was minimal before these incentives and the rising AI-driven electricity demand picture. Maintaining these policy signals is essential to sustaining investment in nuclear energy, ensuring a stable, low-emission power supply for data center expansion.
- **Incentivizing Grid Reliability** – Grid-supporting assets, such as those enabling load balancing and storm resilience, are often not incentivized appropriately in current structures. To ensure these critical capabilities are prioritized in future energy generation, targeted tax or cost incentives will be necessary. Encouraging data center developers to integrate these considerations will be key to building a more resilient and reliable U.S. power grid.
- **Incentivizing Grid Flexibility** – Researchers suggest that increased flexibility in data center operations could enable the existing grid to satisfy gigawatts of incremental demand.⁷ However, a well-designed incentive structure is needed to encourage developers to adopt flexible consumption models while considering local consumer and industrial needs. Further, developers must consider what kinds of compute workloads - AI training, inference, or hybrid cloud activities - will comprise the bulk of their customer demand, as the diversity of said workloads directly influences their hardware and energy needs. Absorbing grid margins may provide only a short- to medium-term solution, underscoring the necessity of new power generation. Clearer guidance and targeted incentives will be essential to ensuring sustainable and strategic energy expansion.

ENABLING EFFECTIVE ADOPTION OF AI BY THE DOMESTIC ENERGY INDUSTRY

The modern economy is built on four essential pillars—steel, cement, plastics, and fertilizer—all of which rely on abundant and affordable natural energy resources. This physical reality directly influences the cost of goods for both consumers and businesses. AI has the potential to improve economic efficiency, but its success depends on reducing costs across industries while enhancing—not replacing—human contributions.

So far, much of the energy sector's AI focus has been on supply-side solutions. However, equal attention must be given to demand-side advancements, which fall into four key areas: (i) accelerating the development of AI-powered tools and data centers that will enhance our nation's ability to identify and safely develop our domestic natural

⁷ [Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems](#)

resources (ii) leveraging AI to enhance productivity across the energy value chain, from extraction, refining and transportation to grid planning, permitting and management (iii) utilizing AI-driven innovations to improve the safety and efficiency in extreme operating conditions and (iv) engaging in AI-powered R&D to drive innovation in new energy sources and materials, ultimately reducing reliance on finite natural resources.

To unlock AI's full potential in the energy industry and accelerate its adoption, we recommend the U.S. to implement the following key policy actions that support these advancements.

- **Infrastructure for Physical AI** – Many of AI's most transformative applications in the energy industry exist in the physical world rather than the virtual one. Unlocking the full potential of Physical AI requires significant infrastructure investments across the data value chain. To accelerate adoption, the government should prioritize R&D funding in key areas, including:
 - **Industrial IoT (IIoT) & Edge Computing** – Deploying ruggedized IIoT sensors with edge computing capabilities to reduce data transmission bottlenecks and enable AI-driven decision-making directly in the field.
 - **Reliable Connectivity in Energy-Rich Regions** – Expanding wired and wireless networks to support real-time data transmission between remote sites and data centers.
 - **Digital Twin Development** – Developing virtual models of physical assets and systems to enhance operational efficiency, predictive maintenance, and risk assessment.
 - **Automation & Robotics** – Advancing infrastructure for autonomous systems to handle repetitive and hazardous tasks in the field such as inspections, repairs, chemical replenishment, and product loading/unloading, ultimately driving down operating expenses across industrial sectors including energy, mining and many others.

Strategic investments in these areas will remove barriers to AI adoption in industrial settings, improving efficiency, safety, and productivity across the energy sector.

- **Physical and Virtual Sandboxes to Accelerate Energy AI Applications** – To drive innovation in AI-powered energy solutions, we propose the creation of physical and virtual sandboxes on public “lands”—dedicated environments where industry leaders, researchers, and AI developers can collaborate free from antitrust concerns, proprietary data restrictions, and cybersecurity risks. These sandboxes would facilitate rapid experimentation and development in several key areas:
 - **High-Performance Computing Access** – Leveraging DOE supercomputers or other national infrastructure to support large-scale AI model training and simulations.

- **Physical Testing Facilities** – Establishing pilot sites for oil & gas production, renewable energy, and next-generation power systems to validate AI applications in real-world conditions.
- **Digital Twin & Virtual Prototyping** – Creating controlled environments for AI-driven modeling, testing, and scenario analysis.
- **AI-Driven Materials Discovery** – Using advanced simulations to accelerate breakthroughs in energy storage, material synthesis, and efficiency improvements.

These sandboxes could be housed within Department of Energy (DOE) Labs, developed in partnership with leading AI firms such as Amazon, or modeled after specialized test facilities funded by the DOE like Colorado State's METEC for methane monitoring. By providing a secure and collaborative ecosystem for experimentation, these centers would serve as proving grounds for breakthrough energy innovations. Establishing these sandboxes on federal lands would also support Secretary Doug Burgum's efforts to unlock our nation's natural resources that are worth trillions.⁸

- **Prizes for AI & Energy** – To drive breakthrough innovation at the intersection of AI and energy, we propose a series of competitions that challenge industry leaders, startups, and researchers to develop cutting-edge solutions. While competitions must be designed carefully, potential competition themes include:
 - **Enhancing Renewable Energy Affordability, Reliability (24/7) and Resiliency** – AI-driven solutions to optimize energy generation, storage, grid stability, and renewable integration.
 - **AI-Designed E&P or Mining Operations** – Building a fully autonomous exploration and production (E&P) or mining capability from the ground up using AI.
 - **AI-Driven Industrial Wastewater Treatment** – Developing intelligent systems for cost-effective and sustainable O&G wastewater management.
 - **Next-Generation Materials Innovation** – Leveraging AI for combinatorial synthesis and modeling to discover new materials with superior performance.

By incentivizing AI-driven breakthroughs in these areas, the U.S. can accelerate advancements in energy, sustainability, and industrial efficiency.

- **Protecting Proprietary Data in AI Applications** – AI's ability to analyze vast multimodal datasets is transformative, but concerns over proprietary and sensitive data often limit access. To encourage broader data sharing while safeguarding privacy and intellectual property, the government should promote:
 - **Advanced Encryption Methods** – Implementing secure encryption techniques to allow AI models to process data without exposing underlying proprietary information.

⁸ [U.S. Secretary Doug Burgum Pitches Resources on Federal Lands as Assets Worth Trillions](#)

- **Anonymization & Differential Privacy** – Developing frameworks that remove or mask sensitive identifiers while preserving data utility for AI-driven insights.
- **Federated Learning & Secure Data Collaboration** – Encouraging AI training methods that enable data analysis without direct access to raw datasets, reducing risks of exposure.

These measures would help mitigate privacy concerns, fostering AI innovation while ensuring that proprietary data remains protected.

- **Ensuring AI Accuracy in the Physical World** – AI accuracy is critical when applied to the physical world, where errors in autonomous navigation, industrial process control, or infrastructure management can have severe consequences. To enhance reliability, the government should promote:
 - **Developing Energy-Specific Compound AI Systems** – Most AI applications in scientifically rigorous fields require deterministic, "classic machine learning (ML)" models to make auditable predictions (e.g. random forest classifiers, multivariate regression, and XGBoost). However, generative AI models' (e.g. transformer models) accessibility enables workers and scientists alike to naturally interface with intelligent systems via simple language. Hybrid, or "compound" AI systems, combine the best of both worlds - an LLM can interpret, challenge, and refine user instructions before executing a classic ML model via function calling or API call, saving users a tremendous amount of time tuning and refining models between runs. This would be a substantial productivity boost across the entire labor class of energy R&D.
 - **Explainability & Auditing** – Requiring transparency in AI decision-making through interpretable models and audit trails to diagnose errors.
 - **Redundant Safety Mechanisms** – Implementing fail-safes and human-in-the-loop oversight in critical applications to mitigate AI failures.
 - **Continuous Learning & Monitoring** – Developing frameworks for ongoing model updates, real-time performance monitoring, and error correction.

By integrating these safeguards, AI systems can achieve the high levels of accuracy and reliability necessary for effective deployment in the physical world.

- **Securing Distributed Energy Resources (DERs) and Microgrids with AI** – As concerns over grid reliability grow, investments in microgrid applications and distributed energy resources (DERs) are expanding. However, with this growth comes heightened cybersecurity risks, as each connected DER asset introduces potential attack vectors, increasing the threat of cyber intrusions, grid manipulation, and cascading failures. To ensure grid security and resilience, the following measures are critical:

- **AI-Driven Threat Detection** – Implementing real-time anomaly detection and predictive analytics to identify and respond to cyber threats before they escalate.
- **Secure Communication Protocols** – Establishing encrypted, standardized frameworks to protect data transmission between DERs, microgrids, and the central grid.
- **Coordinated Incident Response Mechanisms** – Developing AI-assisted security operations centers (SOCs) for rapid response to cyber threats.
- **Regulatory & Compliance Frameworks** – Defining clear standards for data governance, AI transparency, and cybersecurity compliance to prevent vulnerabilities.
- **AI Certification & Testing Standards** – Establishing rigorous AI model testing for critical energy infrastructure, similar to IEC and IEEE certification processes for electrical components.

Standardizing AI applications in the energy sector will be crucial to ensuring safety, reliability, interoperability, and ethical deployment. Developing AI-specific compliance frameworks and risk mitigation strategies will prevent DERs from becoming liabilities while strengthening overall grid security.

- **AI & Workforce Transformation** – AI is set to disrupt nearly 40% of jobs worldwide, transforming industries by both replacing and complementing human roles. In advanced economies, up to 60% of jobs will be impacted, with AI integration enhancing productivity in some cases while reducing labor demand in others⁹. If managed thoughtfully, AI can serve as a force for good, augmenting human capabilities and enabling the workforce to focus on higher-order critical thinking. However, without strategic policies, AI adoption could deepen economic inequality, displace workers, and weaken workforce confidence. To ensure a balanced approach, policymakers must:
 - **Strengthen Workforce Upskilling & Adaptation** – Investing in AI literacy, retraining programs, and education initiatives to help workers transition into AI-enhanced roles.
 - **Preserve Economic Stability & Job Security** – Implementing policies that mitigate job displacement while allowing businesses to leverage AI's efficiencies.
 - **Expand the Skilled Trade Workforce** – The rising power demand for AI will require a surge in electricians, grid technicians, and infrastructure specialists to support data center expansion and grid modernization.

AI should complement, not replace, human labor—driving higher productivity, better wages, and enhanced innovation. A structured human resource infrastructure must be established now to ensure AI adoption strengthens the economy rather than exacerbating job losses.

⁹ IMF report – [AI Will Transform the Global Economy. Let's Make Sure It Benefits Humanity.](#)

DEMOCRATIZING DATA: THE KEY TO AI ADVANCEMENT IN THE ENERGY SECTOR

One of the most impactful government policies to accelerate AI adoption in energy could be democratizing access to existing physical data. In fact, this single action may be the key to unlocking the innovation potential of current AI-enhanced digital assets. In countries like China, the government owns and openly shares¹⁰ vast datasets with the tech industry, driving rapid AI advancements such as DeepSeek and Manus.

In contrast, the U.S. operates under a fragmented data system, where proprietary datasets—especially in the energy sector—are often treated as a competitive “moat”, creating barriers that hinder innovation. Limited access to high-quality physical data slows AI model training, reduces efficiency gains, and restricts technological breakthroughs. Meanwhile, multiple well-funded competitors will double, triple, or quadruple capital spending to get comparable data and resources, making for a much more inefficient use of capital in the aggregate compared to a coordinated approach.

For example, in traditional energy sectors like oil and gas, subsurface geological data provides a major competitive advantage. Companies with access to high-quality geological datasets can significantly reduce exploration risks, optimize drilling strategies, and improve financial performance. However, acquiring this data is often expensive and difficult, especially for smaller companies that lack the resources to purchase proprietary datasets.

The good news is that much of this data exists, but in fragmented, incomplete, or difficult-to-use formats. Consider the following examples:

- Geophysical logs, which measure rock properties such as porosity and resistivity, are legally required to be submitted to state agencies. However, these logs are often incomplete or available only as raster images, making them hard to analyze. A small operator assessing a potential drilling site may have to manually digitize the data—an expensive and time-consuming process. While vendors provide high-quality digitized logs, they come at a high cost, limiting access for smaller players.
- Rock properties and geochemistry data, including porosity, permeability, and hydrocarbon content, are critical for evaluating reservoirs. However, these datasets are often stored in hard-to-use formats like PDFs or paper archives. A geologist studying source rock potential, for instance, may need to manually sift through physical core samples at a state survey office or rely on scattered spreadsheet datasets. While some basic regulatory data (such as oil gravity or

¹⁰ China's policy of “civil-military” fusion in key economic sectors

[China's Evolving Conception of Civil-Military Collaboration](#)

China's 2030 AI development plan, published in 2017, wherein open data sharing and collaboration is mentioned 13 times

[China's 'New Generation Artificial Intelligence Development Plan' \(2017\)](#)

water chemistry) is available, deeper analyses (e.g., SARA fractions) remain proprietary and inaccessible.

- Geophysical data, such as seismic, gravity, and magnetic surveys, is essential for subsurface imaging. While offshore seismic data from USGS is freely available and highly valuable for AI/ML applications in oil, gas, and mineral exploration, onshore seismic data is often prohibitively expensive—costing tens of thousands of dollars or more per square mile. This limits access for smaller companies that could otherwise leverage AI to uncover new energy resources.
- Historical physical records, including maps, books, and field notes housed in geological survey archives and university libraries, contain invaluable observational data. A mining company, for example, might discover century-old field notes detailing ore-bearing formations, but these records are typically available only in physical form, requiring extensive effort to digitize.

These barriers to data accessibility and integration create inefficiencies in exploration, modeling, and decision-making, particularly for smaller operators. Ironically, these smaller firms are often more innovative and agile than their larger counterparts, yet they face the greatest challenges in obtaining the data necessary to compete, coagulating our innovation cycle and national competitiveness. The Shale revolution was started by these scrappy smaller firms which set the stage for the U.S. current dominant position in oil and gas production. These resources were known for decades but only became commercially viable through technological breakthroughs in the last two decades. Providing this data to a broad set of companies, combined with the proposed creation of a physical and virtual sandbox discussed in the prior section, has the potential to extend the U.S. resource production in both oil and gas and mining.

To maintain America's competitive edge in energy and AI and to support Secretary Doug Burgum's vision to map deposits of oil, gas and critical minerals on US federal lands¹¹, we propose the following AI policy actions:

- **Create an Open-Source Subsurface Geological Database Using Publicly Available Data** – Create a standardized digital repository for subsurface geological data across energy-rich basins in the U.S., including both onshore and offshore resources. Ensure data is stored in a machine-readable format with APIs for seamless integration by AI developers and energy companies, enabling real-time analysis and model improvement. The government can build upon the existing publicly available data and enhance the open-source database by offering tax credits or reduced royalty rates from production on public lands.
- **Develop Open-Source Subsurface Models** – Leverage national labs to process the open-source datasets to create AI-driven models with unprecedented insights. Just as META created its open-source Llama model for the tech industry to leverage, the energy industry would similarly benefit from a centrally built

¹¹ [Interior Secretary Doug Burgum touted a plan for mapping deposits of oil, gas and critical minerals on US federal lands](#)

model. These models would be made available to startups and companies, allowing them to refine their own AI-driven models and tools by training on their own proprietary data. This would promote competition and innovation in AI applications and operational execution while preserving proprietary information.

- **Expand Open-Source Databases & Models for AI Development Across the Energy Value Chain** – Establish publicly accessible databases for production, emissions, electricity grid performance, energy consumption, and other energy-related aspects. Enable AI-driven insights into energy optimization, decarbonization, and grid efficiency through structured, well-curated datasets.

By removing barriers to data access, the U.S. can foster AI innovation, empower smaller AI and energy players, and create a lasting competitive advantage in the global energy sector. An open data strategy will not only boost domestic energy efficiency and security but also position the U.S. as a leader in AI-driven energy solutions for the future.

Summary

This AI Action Plan outlines critical policy recommendations to ensure the U.S. has the energy capacity to power AI-driven growth while accelerating AI adoption across the energy sector. We have focused our recommendations on near-term, actionable items that can have an outsized impact on AI and energy strategies which are now inseparably linked—without a resilient, scalable, and affordable energy infrastructure, AI's transformative potential cannot be fully realized.

To maintain global leadership, the U.S. must take decisive action: removing barriers to energy data access, modernizing AI infrastructure, and streamlining permitting to fast-track energy expansion. As China aggressively aligns its AI and energy policies, the U.S. cannot afford a fragmented approach. A coordinated national strategy is essential to avoid a chaotic free-for-all in AI resource deployment—one that could overwhelm an already strained domestic energy grid.

Success will require intra-industry collaboration, bringing together energy producers, AI developers, and policymakers to create an efficient, innovation-driven ecosystem. By aligning AI and energy priorities, fostering cross-sector partnerships, and investing in next-generation infrastructure, the U.S. can turn AI into an engine of industrial renewal—securing both technological and economic dominance in the decades ahead.

Veriten appreciates the Administration's consideration of the above views. We encourage OSTP to contact Robert Kester ([REDACTED]) with any questions or ways that we can assist moving forward.

Sincerely,

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